

Практикум D1

83 marks of the archive, seven techniques, twelve tasks. The smallest topic in the statistics section and the second smallest anywhere in the archive. Fifteen questions in five years, none longer than seven marks, and not one session without one.

О практикуме

Тема взята целиком, и по-другому нельзя: счёт в архиве больше нигде не живёт отдельным вопросом. 89 баллов, 30 блоков — но два из них ноябрьский дубль 2023 года, одна бумага в двух зональных копиях, поэтому рабочих чисел два: 28 вопросов и 83 балла. Самая маленькая тема секции D и вторая по малости во всём архиве после E1.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d1-combinatorics

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D1 — Combinatorics: arrangements, blocks, complements

83 marks of the archive, seven techniques, twelve tasks. The smallest topic in the statistics section and the second smallest anywhere in the archive. Fifteen questions in five years, none longer than seven marks, and not one session without one.

The one idea

Every question in this topic asks the same thing: **how many objects of a certain kind are there?** All the machinery — $n!$, nP_r , nC_r — is three answers to one prior question, and the prior question is the whole topic:

If I build one of these objects, what choices do I make, and does the order of those choices produce a different object?

Answer that and the formula follows. Skip it and no formula helps, because $^{10}P_3$ and $^{10}C_3$ are both perfectly good numbers and only one of them counts what the question is asking about.

That is why the ladder below is not a list of formulas. It is a list of **decisions**, and each rung is the previous one with one more decision made.

How the checks work

They do not know the answers. `verify_count` is handed a description of the objects — how one is built, and what makes it count — and it enumerates them and adds up. The description is the *question* rewritten in Python, not the *method*:

```
verify_count('7a', q7a, product(PENS,
    repeat=5),
            keep=lambda pen: pen[AMBER] !=
            pen[BROWNIE])
```

is *"all the ways of putting five sheep in six pens, keeping the ones where Amber*

and Brownie are apart". It is the sentence from the paper. The method — $6 \times 5 \times 6^3$, or $6^5 - 6^4$ — appears nowhere in it, and that is the part you are here to supply.

When you are wrong the check says **how**, and every diagnosis is built out of the same enumeration:

what you wrote	what the check says
the unrestricted count	the restriction is missing
the count of the bad cases	that is what the question forbids
the answer $\div r!$	an order inside r objects has been dropped
the answer $\times r!$	an order the question does not ask for has been counted
half or twice the answer	the order inside the pair
a fraction	a number of things is a whole number

Two questions have answers that are not counts. *Write down an expression for the number of ways* is checked by `verify_count_law`, which re-enumerates at

small n and compares — so nC_3 , $\frac{n!}{3!(n-3)!}$ and $\frac{n(n-1)(n-2)}{6}$ all pass. *Determine the value of n* is answered as a **set**, **{9}**, and checked by putting it back into the condition. Writing it as a set is not pedantry: the markscheme refuses the last mark if a second value of n is written down beside the right one.

Order of work

level	what it means	tasks
	count the choices; decide whether order matters	1-3
	there is a restriction: glue it, or subtract it	4-7
	one count is not enough	8-12

Every task is a real past-paper question, cited. The full archive of the topic — all twenty-eight questions — is in the companion notebook *D1 archive: combinatorics*.

77% of these marks carry a calculator, and the button really does the arithmetic: nP_r , nC_r and $n!$ are keys,

and 126 126 does not come out in your head. But they are the *last* thing you press. About 28 of the 83 marks are the final A1 of a question; the other 55 are for deciding what to count, and no calculator has an opinion about that.

```
import sys
sys.path.append('..')          # from
practicum/statistics to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
factorial, binomial, permutations, ...

language('en')                 # this
notebook is in English, and so are the checks

n_ = symbols('n')

print('ready; sympy', sp.__version__)
print('a count:      ', factorial(15))
print('a binomial:   ', binomial(9, 6))
print('a set:        ', {9})
print('an enumeration:',
len(list(permutations(range(5), 3))))
```

Map of the seven techniques

#	technique	you recognise it by	it reduces to
1	the product rule	the object is built in steps, and nothing runs out	multiply the choices
2	arrangements	things are placed, each used once	$n!$ or ${}^n P_r$
3	selections	a group is picked and its inside order is nothing	${}^n P_r$ divided by $r!$
4	the block	<i>next to each other, grouped together</i>	glue, arrange, multiply by the inside
5	the complement	<i>not next to each other, at least</i>	all of them, minus technique 4
6	cases	the answer depends on <i>where</i>	techniques 1-5, added
7	a letter in the count	n is asked for, not given	techniques 1-6, then an equation

The first three are one decision made twice. Start with the product rule: $6 \times 6 \times 6$. Now say the objects cannot repeat, and it becomes $6 \times 5 \times 4$ — an arrangement. Now say their order does not matter either, and divide by $3!$ — a selection. Nothing new has been introduced; each rung has taken one more thing away.

Techniques 4 and 5 are the same restriction seen from the two sides. *Together* is a gluing; *apart* is everything minus the gluing. That is why 5 needs 4 and not the other way round.

Techniques 6 and 7 build on all of the above: 6 adds counts that do not overlap, 7 sets one equal to a number and solves for n .

Part 1. Three counts

Theory: one decision, made twice

Build one object. You make a sequence of choices. If the choices are independent, the counts multiply — that is the whole of the **product rule**, and it is the only genuinely primitive idea in the topic:

$$\begin{aligned}\text{a three-symbol code from eleven symbols} &= 11 \times 11 \times 11 \\ &= 1331\end{aligned}$$

Now add a condition: **no symbol is used twice**. Each choice has one option fewer than the one before it:

$$11 \times 10 \times 9 = 990 = \frac{11!}{8!} = {}^{11}P_3$$

That is an **arrangement**. nP_r is not a new idea; it is the product rule with the supply running down.

Now add a second condition: **the order of the chosen symbols does not matter**. Every group of three has been counted once for each of its $3!$ orderings, so:

$${}^{11}C_3 = \frac{{}^{11}P_3}{3!} = \frac{990}{6} = 165$$

A **selection**. Again nothing new — a division by an over-count.

Three formulas, one idea, two decisions: *can things repeat?* and *does their order make a different object?* Answer those two and you have already chosen the formula.

The decision is not always announced

Sometimes the paper says *order is not important*. More often it does not, and the giveaway is subtler. Consider four-letter codes made from the eight letters A–H, all different, with the letters in **alphabetical order**. Order matters enormously to the code — ABDG and GDBA are different strings — but you do not get to choose it: each set of four letters produces exactly **one** alphabetical code. So the count is a selection, ${}^8C_4 = 70$, and a markscheme gives a whole mark

for saying why. The sentence it wants has this shape:

"every unordered subset can be arranged in exactly one way into a string with the letters in increasing order."

Whenever a condition pins the order down completely, the arrangements collapse to selections. Look for the words *increasing*, *decreasing*, *alphabetical*, *in size order*.

Restricted positions go first

If one position is special, fill it first. Seven people in a row, one of whom must take one of the two end seats:

$$\underbrace{2}_{\text{the two ends}} \times \underbrace{6!}_{\text{everyone else}} = 2 \times 720 = 1440$$

Markschemes for such questions name the wrong answers explicitly, and they all come from filling the special position last: one arrangement count that ignored the restriction altogether, and another that applied it to the wrong pool of

objects. Both look like plausible permutations, which is exactly why they are named.

Task 1 ● — *November 2022 Paper 1 Q11(a), 4 marks*

Consider a three-digit code abc , where each of a , b and c is assigned one of the values 1, 2, 3, 4 or 5.

Find the total number of possible codes

(a) assuming that each value can be repeated (for example, 121 or 444); **(b)** assuming that no value is repeated.

```
VALUES = (1, 2, 3, 4, 5)
```

```
q1a = ...
```

```
q1b = ...
```

```
verify_count('1a', q1a, product(VALUES,  
repeat=3))
```

```
verify_count('1b', q1b, permutations(VALUES,  
3))
```

Task 2 ● — *May 2022 TZ2 Paper 2 Q9, 4 marks*

Consider the set of six-digit positive integers that can be formed from the

digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

Find the total number of six-digit positive integers that can be formed such that

(a) the digits are distinct; **(b)** the digits are distinct and are in increasing order.

Both parts count the same objects. Only the second sentence differs, and it is worth thinking about what it actually removes.

```
# a six-digit number with distinct digits is
six different digits in places
q2a = ...
q2b = ...

verify_count('2a', q2a,
permutations(range(10), 6),
              keep=lambda digits: digits[0] !=
0)
verify_count('2b', q2b,
permutations(range(10), 6),
              keep=lambda digits: digits[0] !=
0 and list(digits) == sorted(digits))
```

Task 3 — November 2025 TZ1 **Paper 2 Q2, 4 marks**

A particular service group at a school has ten members.

The group intends to select a committee of seven by first choosing a chairperson, a vice-chairperson and a treasurer, then selecting four additional members to complete the committee.

Determine the number of ways in which this committee can be chosen.

One question, both decisions, in that order. The three officers are distinguishable from one another; the four extra members are not.

```
MEMBERS = range(10)

# a committee: three named posts, then four
ordinary members
def committees():
    for officers in permutations(MEMBERS, 3):
        rest = [m for m in MEMBERS if m not
in officers]
        for others in combinations(rest, 4):
            yield officers, others

q3 = ...

verify_count('3', q3, committees())
```

Part 2. One restriction

Theory: glue it, or subtract it

Almost every restriction in this topic is one of two sentences:

- **together** — *next to each other, immediately after, grouped;*
- **apart** — *not next to each other, must not share, at least one seat away.*

They are not two techniques. They are one technique and its complement.

Together: glue and count the inside

Treat the things that must stand together as **one object**, arrange what you now have, and multiply by the orders **inside** the glued object.

Six people in a line with two of them together:

$$\begin{array}{ccccc} \underbrace{5!} & \times & \underbrace{2!} & = & 120 \times 2 \\ \text{four others + the block} & & \text{inside the block} & & \\ & & & & = 240 \end{array}$$

The $2!$ is the mark people lose. Markschemes award M1 for *"an attempt to consider the pair as a single object"* and A1 for the product — the second mark is entirely the inside, and it grows fast: a block of three carries $3! = 6$, a block of four carries 24.

Two glued objects means two insides and one outside. Seven books in three groups — three, two and two — each group kept together:

$$\underbrace{3! \times 2! \times 2!}_{\text{inside each group}} \times \underbrace{3!}_{\text{the order of the groups}} = 24 \times 6 = 144$$

and the last $3!$ gets its own M1: *"recognise that the three groups can be placed in any order."* Forgetting it is the commonest single error of the rung, because the groups look like fixtures rather than objects.

When the block does not fill the row, count its positions. Three people sitting together somewhere in a row of

seven seats: the block can start at seat 1 through seat 5, so

$$5 \times 3! = 30$$

Five positions, not $5!$ objects. Count where the block can begin — that is $(\text{seats} - \text{block} + 1)$ — and multiply by the orders inside it.

Apart: count everything, then remove the together

The direct count of *apart* splits into cases and takes ten minutes. The complement takes two lines:

$$\text{apart} = \text{all} - \text{together}$$

and you already know how to count *together*. Five people in a row, two particular ones **not** next to each other:

$$5! - \underbrace{2! \times 4!}_{\text{glued, then arranged}} = 120 - 48 = 72$$

In a row the *together* count is the glue formula and nothing more. On a board or a grid it is not: there you have to count the adjacent **pairs** from the picture,

because how many of them there are depends on the shape, not on the number of cells. Counting the pairs wrong is the one place this method fails, and it fails silently.

Which whole are you subtracting from? May 2024 TZ1 asks for arrangements with Alvin and Bobby adjacent, then asks again with Catalina and Daniela *also* forbidden from being adjacent. The second answer is not 10! minus something. Alvin and Bobby are still glued, so the whole is the answer to part (a).

Symmetry is the complement in disguise

One runner finishes somewhere after another. Every finishing order has a mirror image in which the two are swapped, and exactly one of the pair has the first runner later. So the answer is half of everything — for six runners,

$$\frac{6!}{2} = 360.$$

No cases, no gluing. It works whenever the condition and its negation are in bijection — and it is worth spotting, because the case-by-case route here is $(5 + 4 + 3 + 2 + 1) \times 4!$ and takes several times as long for the same number.

Task 4 — **November 2025 TZ3**
Paper 2 Q8, 7 marks

Malik owns 15 recipe books which he bought while travelling.

(a) Find the number of different ways of arranging the books in a line on a shelf.

Malik's books were bought in three continents. Six books are from Asia, five from Europe and four from Africa. Malik decides to arrange the books on the shelf so that the books from each of the continents are grouped together.

(b) Determine the number of different ways that Malik can do this.

Malik chooses four books, all from the same continent.

(c) Determine the number of different choices Malik could make.

```

BOOKS = tuple('A' * 6 + 'E' * 5 + 'F' * 4)
# each book carries its continent

# the books of each continent stand on the
shelf in one run
def in_one_run(shelf):
    line = ''.join(shelf)
    return all(c * line.count(c) in line for
c in 'AEF')

q4a = ...
q4b = ...
q4c = ...

# 15! arrangements will not fit; the first
four places are enumerated and
# the remaining eleven books contribute a
factor of 11!
verify_count('4a', q4a,
permutations(range(15), 4),
each=factorial(11))
# here the shelf pattern is enumerated –
which continent stands where –
# and each pattern stands for 6!*5!*4!
arrangements of the books themselves
verify_count('4b', q4b,
multiset_permutations(list(BOOKS)),
keep=in_one_run,
                each=factorial(6) * factorial(5)
* factorial(4))
verify_count('4c', q4c, combinations(BOOKS,
4),
                keep=lambda four: len(set(four))
== 1)

```

Task 5 — May 2021 TZ2 Paper 2

Q7, 5 marks

Eight runners compete in a race where there are no tied finishes. Andrea and Jack are two of the eight competitors in this race.

Find the total number of possible ways in which the eight runners can finish if Jack finishes

(a) in the position immediately after Andrea; **(b)** in any position after Andrea.

Part (a) is a block with one inside order, not two. Part (b) is not a block at all.

```
ANDREA, JACK = 0, 1
RUNNERS = range(8)

q5a = ...
q5b = ...

verify_count('5a', q5a,
permutations(RUNNERS),
            keep=lambda order:
order.index(JACK) == order.index(ANDREA) + 1)
verify_count('5b', q5b,
permutations(RUNNERS),
            keep=lambda order:
order.index(JACK) > order.index(ANDREA))
```

Task 6 ● — *May 2022 TZ1 Paper 2*
Q9, 7 marks

Mary, three female friends, and her brother, Peter, attend the theatre. In the theatre there is a row of 10 empty seats. For the first half of the show, they decide to sit next to each other in this row.

(a) Find the number of ways these five people can be seated in this row.

For the second half of the show, they return to the same row of 10 empty seats. The four girls decide to sit at least one seat apart from Peter. The four girls do not have to sit next to each other.

(b) Find the number of ways these five people can now be seated in this row.

```

PETER, GIRLS = 0, (1, 2, 3, 4)
SEATS = range(10)

q6a = ...
q6b = ...

# seat[i] is the seat that person i takes
verify_count('6a', q6a, permutations(SEATS,
5),
            keep=lambda seat: max(seat) -
min(seat) == 4)
verify_count('6b', q6b, permutations(SEATS,
5),
            keep=lambda seat:
all(abs(seat[g] - seat[PETER]) > 1 for g in
GIRLS))

```

Task 7 — May 2021 TZ1 Paper 1 Q9, 8 marks

A farmer has six sheep pens, arranged in a grid with **three rows and two columns**. Five sheep called Amber, Brownie, Curly, Daisy and Eden are to be placed in the pens. Amber and Brownie are known to fight.

Find the number of ways of placing the sheep in the pens in each of the following cases:

(a) Each pen is large enough to contain five sheep. Amber and Brownie must not be placed in the same pen. **(b)** Each pen may only contain one sheep. Amber and Brownie must not be placed in pens which share a boundary.

Paper 1, so no calculator, and none is needed. Part (a) is the product rule with one thing forbidden; part (b) is the complement, and its whole difficulty is counting the boundaries.

```

AMBER, BROWNIE = 0, 1
PENS = range(6)                                # pen p sits
in row p // 2, column p % 2

# pens sharing a side in the three-by-two
grid
def share_a_boundary(one, other):
    return abs(one // 2 - other // 2) +
abs(one % 2 - other % 2) == 1

q7a = ...
q7b = ...

# (a) a pen holds every sheep: each sheep
picks a pen, and pens do not fill up
verify_count('7a', q7a, product(PENS,
repeat=5),
            keep=lambda pen: pen[AMBER] !=
pen[BROWNIE])
# (b) one sheep to a pen
verify_count('7b', q7b, permutations(PENS,
5),
            keep=lambda pen: not
share_a_boundary(pen[AMBER], pen[BROWNIE]))

```

Part 3. When one count is not enough

Theory: split, add, and solve

Cases

Sometimes no single expression covers the question, because the number of options depends on **how the first choice went**. A committee of four from five men and four women, with **at least two women**, splits by how many women there are:

- two women, two men: ${}^4C_2 \times {}^5C_2 = 6 \times 10 = 60$;
- three women, one man: ${}^4C_3 \times {}^5C_1 = 4 \times 5 = 20$;
- four women: ${}^4C_4 = 1$.

$$60 + 20 + 1 = 81$$

Two rules govern every case split, and both are worth checking out loud:

They must not overlap. Ask: can one object land in two cases at once? If it

can, you have counted it twice. Here a committee has one definite number of women, so it cannot.

They must cover everything. The last term above is the one people drop: there is only one way to take all four women, it produces a single committee, and it does not *feel* like a case. Markschemes give a separate M1 just for *"recognition that the sum of 3 different cases is required."*

The split does not have to be by count. For pairs from $\{1, \dots, 20\}$ whose **sum is even**, sort the numbers by parity — ten odd, ten even — and a pair works exactly when both come from the same half:

$${}^{10}C_2 + {}^{10}C_2 = 45 + 45 = 90$$

Choosing what to split by is the whole of the technique. Position, count, parity, remainder: pick the property that makes the options inside each case stop depending on anything else.

Identical groups: divide by their order

Eight people into two groups of four. Choosing group A and letting the rest be group B gives

$${}^8C_4 \times {}^4C_4 = 70 \times 1 = 70$$

but the groups have no names. Each actual split has been counted once for every order in which its two groups could have been picked — $2!$ of them:

$$\frac{70}{2!} = 35$$

There is a way to avoid the division entirely, and markschemes accept it: put a fixed person in a group and choose their three companions from the seven who are left; whoever remains forms the other group by itself:

$${}^7C_3 = 35$$

No over-count happens, so no division is needed. With g identical groups the division is by $g!$ — and the fixed-person trick removes it by making one group

distinguishable, which is the same idea seen from the other side. The two methods disagree about nothing except how much you have to remember.

A letter in the count

The last rung is every rung before it, with n where a number used to be. Three things go wrong here and all three are worth naming.

Cancel, do not expand. From ${}^nC_3 = 4 {}^nC_2$ write both sides out as products and strike out what is common:

$$\begin{aligned}\frac{n(n-1)(n-2)}{6} &= 4 \cdot \frac{n(n-1)}{2} \\ &\implies \frac{n-2}{6} \\ &= 2 \\ &\implies n \\ &= 14.\end{aligned}$$

The $n(n-1)$ went away without ever being multiplied out. Expanding $n!$ gets you nowhere; markschemes award M1 for *"a valid attempt to eliminate all factorials."*

Throw away the root the question forbids. When the cancellation leaves a quadratic there are two roots and usually one of them is impossible. ${}^nC_2 = 45$ gives $n(n-1) = 90$, so $n = 10$ or $n = -9$, and n counts objects. Elsewhere the second root is positive but still forbidden — a group would come out too small, a case would be empty — and only the question says so. Markschemes are blunt about it: *"Do not award the final A1 if additional values of n are given."* The rejection is a mark, not a tidy-up, which is why answers of this kind are written as a set.

Round an inequality towards the condition, not towards the arithmetic. $\frac{6}{n(n-1)} > 0.05$ gives $n < 11.47$, so the greatest integer is 11. The markscheme also accepts the table: $n = 11$ gives $0.0545\dots$ and $n = 12$ gives $0.0454\dots$ — checking both sides of the boundary *is* the solution, and it earns the same marks.

Task 8 ● — November 2023 TZ1
Paper 2 Q7, 6 marks

A junior baseball team consists of six boys and three girls. The team members are to be placed in a line to have their photograph taken.

(a) In how many ways can the team members be placed if **(i)** there are no restrictions; **(ii)** the girls must be placed next to each other?

(b) Five members of the team are selected to attend a baseball summer camp. Find the number of possible selections that contain at least two girls.

Part (b) has two honest routes and one dishonest one. Add the two cases that qualify, or subtract the two that do not, from everything. Mixing the halves of the two routes gives a plausible number and a wrong one.

```

TEAM = range(9)
GIRLS = {0, 1, 2}                                # the six
boys are 3..8

def girls_in(line):
    return [i for i, member in
enumerate(line) if member in GIRLS]

q8a = ...
q8b = ...
q8c = ...

verify_count('8a', q8a, permutations(Team))
verify_count('8b', q8b, permutations(Team),
              keep=lambda line:
max(girls_in(line)) - min(girls_in(line)) ==
2)
verify_count('8c', q8c, combinations(Team,
5),
              keep=lambda five: len(GIRLS &
set(five)) >= 2)

```

Task 9 — May 2024 TZ1 Paper 2 Q9, 7 marks

A group of 10 children includes one pair of brothers, Alvin and Bobby, and one pair of sisters, Catalina and Daniela. The children are to be seated at 10 desks arranged in **two rows of five**.

Alvin and Bobby must be seated next to each other **in the same row**.

(a) Find the total number of ways the children can be seated.

After an argument, Catalina and Daniela must not be seated next to each other. Alvin and Bobby must still be seated next to each other.

(b) Find the total number of ways the children can be seated.

"In the same row" is the whole question. Desks 5 and 6 are next to each other on the page and not next to each other in the problem.

```

ALVIN, BOBBY, CATALINA, DANIELA = 0, 1, 2, 3
DESKS = range(10)                # 0-4 front
row, 5-9 back row, left to right

# neighbouring desks in the same row
def side_by_side(one, other):
    return one // 5 == other // 5 and abs(one
- other) == 1

q9a = ...
q9b = ...

# the restriction touches only the named
children; the rest give a factor
verify_count('9a', q9a, permutations(DESKS,
2),
            keep=lambda desk:
side_by_side(desk[ALVIN], desk[BOBBY]),
            each=factorial(8))
verify_count('9b', q9b, permutations(DESKS,
4),
            keep=lambda desk:
side_by_side(desk[ALVIN], desk[BOBBY])
and not
side_by_side(desk[CATALINA], desk[DANIELA]),
            each=factorial(6))

```

Task 10 — May 2023 TZ2 Paper 2 Q9, 5 marks

Let S be the set of 30 positive integers $\{1, 2, 3, \dots, 28, 29, 30\}$.

Raghu randomly selects three positive integers from S without replacement. He

then adds them together and determines whether the sum is divisible by 3.

Determine the total number of selections Raghu can make to obtain a sum that is divisible by 3.

You may assume that order is not important, for example, $\{1, 2, 3\}$, $\{1, 3, 2\}$, $\{2, 3, 1\}$, $\{2, 1, 3\}$, $\{3, 1, 2\}$, $\{3, 2, 1\}$ are all considered to be the same selection.

Thirty numbers, but only three kinds of number. Sorting them by remainder turns this into a small question.

```
q10 = ...  
  
verify_count('10', q10, combinations(range(1,  
31), 3),  
keep=lambda three: sum(three) %  
3 == 0)
```

Task 11 — May 2025 TZ2 Paper 2 Q11(a), 3 marks

A mathematics class of 15 students plays a game which requires three equal size teams.

Find the total number of ways that the three teams can be chosen.

The teams have no names. Whatever you compute first, ask whether the same three teams could have been produced twice.

```
CLASS = range(15)

# splits of the class into three fives; the
# teams have no names, so the
# same three fives written in another order
# are the same split
def team_splits():
    seen = set()
    for first in combinations(CLASS, 5):
        rest = [s for s in CLASS if s not in
first]
        for second in combinations(rest, 5):
            third = tuple(s for s in rest if
s not in second)
            split = frozenset((first, second,
third))
            if split not in seen:
                seen.add(split)
                yield split

q11 = ...

verify_count('11', q11, team_splits())
# the enumeration takes a second or two
```

Task 12 ● — May 2024 TZ2 Paper 1
Q9, 7 marks

A teacher takes n students on a field trip. The students are assigned randomly into two groups. For safety reasons there must be exactly three students in the first group and at least three students in the second group. The teacher will randomly assign three students to the first group and the other students to the second group.

(a) Write down an expression for the number of ways that the students could be assigned.

Two of the students ask the teacher not to work in the same group. The teacher agrees and now finds that the number of ways to assign the students is halved.

(b) Determine the value of n .

Part (b) is answered as a set, {9}. Both roots of the quadratic satisfy the equation; only one of them satisfies the question.

```

# how many ways to pick three out of size –
# and, if apart, so that the two
# named students land in different groups
def assignments(size, apart=False):
    groups = combinations(range(size), 3)
    if apart:
        groups = (g for g in groups if
len({0, 1} & set(g)) == 1)
    return sum(1 for _ in groups)

# the condition of part (b), read literally
def halved(size):
    if not sympify(size).is_integer or size <
4:
        return None
# a number of students is a whole number
    size = int(size)
    return size - 3 >= 3 and 2 *
assignments(size, apart=True) ==
assignments(size)

q12a = ...
q12b = {...}

verify_count_law('12a', q12a, n_,
assignments, sizes=(5, 6, 7, 8, 9))
verify_param_set('12b', q12b, halved, var=n_,
window=(4, 30))

```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**. On the paper that decision takes about five seconds, and everything else follows from it.

code	technique
product	steps multiply; nothing runs out
perm	things are placed, each used once
comb	a group is chosen and its inside order is nothing
block	something must stay together: glue it
minus	something must stay apart: count all, subtract the together
cases	it depends on where; split and add
letter	n is what is asked for

1. A three-digit code is made from the values 1 to 5, each of which may be repeated. How many codes are there?
2. Six-digit numbers are made from distinct digits and cannot start with 0. How many are there?

3. Three integers are chosen from thirty; order is not important. How many selections are there altogether?
4. Fifteen books are shelved so that the books from each continent stand together. How many arrangements?
5. Alvin and Bobby sit next to each other; Catalina and Daniela must not. How many seatings?
6. Three integers are chosen from thirty. How many of those selections have a sum divisible by 3?
7. The number of ways is halved when two students are separated. Find the number of students.
8. Jack finishes in the position immediately after Andrea. How many finishing orders?
9. Amber and Brownie must not be placed in pens which share a boundary. How many placements?
0. Fifteen students form three equal teams with no names. How many ways?

1. Four books are chosen, all from the same continent. How many choices?
2. A chairperson, a vice-chairperson and a treasurer are chosen from ten members. How many ways?

```
answers = {  
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',  
    7: '', 8: '', 9: '', 10: '', 11: '', 12:  
    '',  
}  
  
trigger_check(answers, {1: 'a8792157cb4f', 2:  
'017a80b680ac', 3: 'ef4e3f507983', 4:  
'496aca80e4d8', 5: '129fa7713adb', 6:  
'352b84777d8d', 7: 'f80105efab6d', 8:  
'496aca80e4d8', 9: '129fa7713adb', 10:  
'ef4e3f507983', 11: '352b84777d8d', 12:  
'017a80b680ac'})
```

On the clock — *May 2025 TZ3* ***Paper 2 Q8, 5 marks***

Five marks, seven minutes. No hints this time.

A class of students plays a tic-tac-toe competition among themselves. Each

individual game in the competition involves only two students.

Every student in the class is to play every other student **twice**. However, Stephen left the class after he had played only seven games. All other games, not involving Stephen, were played.

By the end of the competition a total of 513 games had been played.

Determine the number of students that were originally in the class.

Answer as a set, as in task 12.

```
# whether a class of size students played
exactly 513 games
def games(size):
    if not sympify(size).is_integer or size <
3:
        return None
    size = int(size)
    everyone = sum(2 for _ in
combinations(range(size), 2))    # everyone
plays everyone twice
    missed = 2 * (size - 1) - 7
# Stephen's games that were not played
    return everyone - missed == 513

qt = {...}

verify_param_set('timer', qt, games, var=n_,
window=(3, 40))
```



Solutions

1 (a) Three independent choices out of five: $5 \times 5 \times 5 = \boxed{125}$.

1 (b) The supply runs down: $5 \times 4 \times 3 = {}^5P_3 = \boxed{60}$.

The two parts are printed side by side in the paper for a reason. The only difference between them is one sentence, and it changes 5^3 into $\frac{5!}{2!}$.

2 (a) Fill the restricted position first. The leading digit has 9 choices (anything but 0); the other five positions take five of the nine digits that are left, in order:

$$9 \times {}^9P_5 = 9 \times 15120 = \boxed{136080}$$

${}^{10}P_6 = 151200$ ignores the restriction; ${}^9P_6 = 60480$ applies it and then forgets that 0 is available for the later positions. Both are named in the markscheme, and both score M1A0.

2 (b) An increasing six-digit number cannot contain 0 — it would have to come first, and it cannot. So choose six digits from $\{1, \dots, 9\}$; each choice can be written in increasing order in exactly one way:

$${}^9C_6 \times 1 = \boxed{84}$$

Equivalently 9C_3 : choosing which three digits to leave out.

3 The three officers are distinguishable, so their selection is an arrangement; the four extra members are not, so theirs is a selection:

$${}^{10}P_3 \times {}^7C_4 = 720 \times 35 = \boxed{25200}$$

${}^{10}C_3 \times 3!$ is the same number written the other way round, and the markscheme accepts it. What it does not accept is ${}^{10}C_3 \times {}^7C_4$ — that is 4200, and it has thrown away the fact that a chairperson is not a treasurer.

4 (a) Fifteen distinct books in a line:
 $\boxed{15!} = 1.31 \times 10^{12}.$

The markscheme accepts either form.
There is no reason to evaluate it.

4 (b) Three blocks. Arrange inside each, then arrange the blocks:

$$6! \times 5! \times 4! \times 3! = 720 \times 120 \times 24 \times 6 \\ = \boxed{12441600}$$

The final $3!$ is the mark most often lost.
Asia, Europe, Africa is one shelf; Africa, Asia, Europe is another.

4 (c) Three cases, one per continent, and they cannot overlap:

$${}^6C_4 + {}^5C_4 + {}^4C_4 = 15 + 5 + 1 = \boxed{21}$$

The ${}^4C_4 = 1$ is the term that disappears.
There *is* one way to take all four African books, and it is worth a mark.

5 (a) *Immediately after* glues Andrea and Jack into one object with **one**

possible inside order — Andrea first. So there are seven objects:

$$7! = \boxed{5040}$$

Multiplying by $2!$ here is the standard slip, and it doubles the answer. The block technique multiplies by the number of orders *the question allows*, which is not always the number of orders that exist.

5 (b) For every finishing order, swapping Andrea and Jack gives another one, and exactly one of the pair has Jack later. So it is half of everything:

$$\frac{8!}{2} = \frac{40320}{2} = \boxed{20160}$$

The markscheme's second method counts positions: $(7 + 6 + 5 + 4 + 3 + 2 + 1) \times 6! = 28 \times 720 = 20160$. Same answer, seven times the work.

6 (a) The five sit as a block somewhere in a row of ten. The block occupies five consecutive seats, so it can start at seat 1, 2, 3, 4, 5 or 6 — six positions — and

the five people can be arranged inside it in $5!$ ways:

$$6 \times 5! = 6 \times 120 = \boxed{720}$$

The markscheme adds "*accept 6!*", because $6 \times 5!$ happens to equal $6!$. It does not mean the block is a sixth object; try the same question in a row of eleven seats and the two numbers separate.

6 (b) Peter's neighbours must be empty, and how many neighbours he has depends on where he sits:

- Peter at either end — 2 choices. One seat next to him is barred, leaving 8 seats for the four girls, in order: $2 \times {}^8P_4 = 2 \times 1680 = 3360$.
- Peter not at an end — 8 choices. Two seats are barred, leaving 7: $8 \times {}^7P_4 = 8 \times 840 = 6720$.

$$3360 + 6720 = \boxed{10080}$$

The cases do not overlap (Peter is either at an end or not) and they cover everything (there is no third possibility).

Both facts are worth stating before you add.

7 (a) Every pen holds all five sheep, so each sheep chooses a pen freely — except that Brownie cannot repeat Amber's:

$$\underbrace{6}_{\text{Amber}} \times \underbrace{5}_{\text{Brownie}} \times \underbrace{6^3}_{\text{Curly, Daisy, Eden}} = 5 \times 6^4 = \boxed{6480}$$

Or through the complement: $6^5 - 6^4 = 7776 - 1296 = 6480$. Both are in the markscheme, and the second is a preview of part (b).

7 (b) One sheep per pen, so this is ${}^6P_5 = 6! = 720$ placements in all. Now count the forbidden ones. In a grid of three rows and two columns the pens that share a boundary are:

$$\underbrace{3}_{\text{horizontal pairs}} + \underbrace{4}_{\text{vertical pairs}} = 7$$

For each of the 7 pairs, Amber and Brownie can go two ways round, and the

other three sheep fill three of the four remaining pens in $4!$ ordered ways:

$$720 - 2 \times 7 \times 4! = 720 - 336 = \boxed{384}$$

The markscheme's second method is the case split: Amber in a corner pen (4 of them, 3 pens then free for Brownie) or in a middle pen (2 of them, 2 free), giving $4 \times 3 \times 4! + 2 \times 2 \times 4! = (12 + 4) \times 24 = 384$.

Corner pens have two neighbours, middle pens have three — which is the same observation as the 7, seen from the other side.

8 (a)(i) Nine people, no restrictions: $9! = \boxed{362880}$.

8 (a)(ii) The three girls become one object, so seven objects are arranged, and the girls have $3!$ orders inside:

$$7! \times 3! = 5040 \times 6 = \boxed{30240}$$

8 (b) *At least two girls* out of three, in a selection of five from nine. Two routes:

$$\underbrace{{}^6C_3 \times {}^3C_2}_{\text{exactly 2 girls}} + \underbrace{{}^6C_2 \times {}^3C_3}_{\text{exactly 3 girls}} = 60 + 15 = \boxed{75}$$

or, from the other end,

$$\begin{aligned} & \underbrace{{}^9C_5}_{\text{all}} \\ & - \underbrace{{}^3C_1 \times {}^6C_4}_{\text{one girl}} \\ & - \underbrace{{}^6C_5}_{\text{no girls}} \qquad = 126 \\ & - 45 \\ & - 6 \\ & = 75 \end{aligned}$$

Both are in the markscheme. The number of girls is at most three, so "at least two" is only two cases — which is why adding is as short as subtracting here. With six girls it would not be.

9 (a) *Next to each other in the same row.* Each row of five desks has four adjacent pairs, so there are $2 \times 4 = 8$ pairs of desks, and Alvin and Bobby can sit in either order:

$$\begin{array}{rcl}
 \underbrace{8 \times 2}_{=16} \times & \underbrace{8!}_{\text{the other eight children}} & = 16 \times 40320 \\
 & & = \boxed{645120}
 \end{array}$$

The markscheme's second method is worth seeing: treat the ten desks as one line of ten, which has nine adjacent pairs, then remove the pair that straddles the two rows — $2 \times 9! - 2 \times 8! = 725760 - 80640 = 645120$.

9 (b) Alvin and Bobby stay glued, so the whole to subtract from is part (a), not $10!$. Count the arrangements where Catalina and Daniela are *also* adjacent, splitting on where Alvin and Bobby sit:

- Alvin and Bobby at the end of a row — 8 ways (2 rows $\times$ 2 ends $\times$ 2 orders). Three desks are left in that row, giving 2 adjacent pairs, plus 4 in the other row: 6 pairs, 2 orders, so $8 \times 12 \times 6! = 69120$.
- Alvin and Bobby not at an end — also 8 ways. Now the leftover desks in their row give only 1 adjacent pair, plus 4: 5 pairs, 2 orders, so $8 \times 10 \times 6! = 57600$.

$$645120 - (69120 + 57600) = \boxed{518400}$$

10 Every integer leaves remainder 0, 1 or 2 on division by 3, and $\{1, \dots, 30\}$ splits into three groups of exactly ten. The sum of three numbers is divisible by 3 exactly when their remainders are all the same or all different:

$$\begin{aligned}
 & \underbrace{3 \times {}^{10}C_3}_{\text{all from one class}} \\
 & + \underbrace{10 \times 10 \times 10}_{\text{one from each}} = 360 \\
 & + 1000 \\
 & = \boxed{1360}
 \end{aligned}$$

The complement works too and the markscheme gives it: ${}^{30}C_3 = 4060$ in all, and the selections whose sum is *not* divisible by 3 come to 2700.

11 Choosing the teams one after another counts each split 3! times:

$$\begin{aligned}\frac{{}^{15}C_5 \times {}^{10}C_5 \times {}^5C_5}{3!} &= \frac{3003 \times 252 \times 1}{6} \\ &= \frac{756756}{6} \\ &= \boxed{126126}\end{aligned}$$

Equivalently $\frac{15!}{5! 5! 5! 3!}$.

The markscheme's second method avoids the division: fix one student, choose their four teammates from the other fourteen, then fix any student not yet placed and choose theirs from the nine left:

$$\begin{aligned}{}^{14}C_4 \times {}^9C_4 \times {}^4C_4 &= 1001 \times 126 \times 1 \\ &= 126126\end{aligned}$$

Nobody is over-counted, so nobody has to be divided out. Both are three marks; the second is harder to get wrong.

12 (a) Three students out of n , order irrelevant, and the rest are the second group: $\boxed{{}^nC_3}$.

12 (b) With the two students separated: choose which of them goes into the first

group (${}^2C_1 = 2$), then two more from the other $n - 2$:

$$\begin{aligned} {}^2C_1 \times {}^{n-2}C_2 &= 2 \times \frac{(n-2)(n-3)}{2} \\ &= (n-2)(n-3) \end{aligned}$$

This is half of part (a):

$$\frac{1}{2} \cdot \frac{n(n-1)(n-2)}{6} = (n-2)(n-3)$$

Cancel $(n-2)$ — legitimate, since $n \geq 6$ — and multiply by 12:

$$\begin{aligned} n(n-1) &= 12(n-3) \\ \implies n^2 - 13n + 36 & \\ &= 0 \\ \implies (n-9)(n-4) & \\ &= 0 \end{aligned}$$

$n = 4$ satisfies the equation. It does not satisfy the question: with four students the second group would hold one, and the trip requires at least three. So $n = \boxed{\{9\}}$.

The markscheme does not spell the rejection out; it just writes *"Do not award the final A1 if additional values of n are*

given." One of the six marks is for noticing that the algebra offered you something the situation cannot use.

Timer

Everyone plays everyone twice, so a class of n plays $2 \times {}^nC_2 = n(n-1)$ games. Stephen should have played $2(n-1)$ and played 7, so $2(n-1) - 7$ games are missing:

$$n(n-1) - (2(n-1) - 7) = 513$$

$$n^2 - n - 2n + 2 + 7 = 513$$

$$\implies n^2 - 3n - 504$$

$$= 0$$

$$\implies (n-24)(n+21)$$

$$= 0$$

$n = -21$ is not a number of students, so $n = \boxed{\{24\}}$.

Check: $24 \times 23 = 552$ games if nobody had left, and $2 \times 23 - 7 = 39$ of them were not played. $552 - 39 = 513$.

The markscheme's other route counts the $N = n - 1$ students who are not Stephen: they play $N(N - 1)$ games among themselves, plus Stephen's 7, giving $N(N - 1) + 7 = 513$, $N = 23$, and one more for Stephen.